

> customer_data

	Age	visits	Gender	Membership	Purchase
1	22	2	male	Regular	Low
2	35	5	Female	Gold	High
3	40	6	Female	Gold	High
4	28	3	male	Regular	Low
5	50	8	Female	Gold	High
6	45	7	male	Silver	High
7	30	4	Female	Silver	High
8	55	9	male	Gold	Medium
9	27	3	male	Regular	Low
10	38	6	Female	Silver	Medium

Ex: No: 6A
Date: 21/4/26

Customer Purchase Category Prediction Using Naive Bayes Classifier

Problem Description:

An e-commerce company wants to predict whether a customer will make a **High** or **Low** purchase based on demographic and behavioral attributes. The dataset contains both categorical (Gender, Membership) and numerical (Age, Visits per Month) features. To perform probabilistic classification and handle mixed data types efficiently, the **Naive Bayes** classifier is used.

Objective:

To implement the Naive Bayes classification algorithm in R by training a model on customer data, predicting purchase categories, and evaluating classification accuracy.

Procedure:

- Step 1: Create Dataset - customer_data

Age	Visits	Gender	Membership	Purchase
1	22	2	Male	Regular
2	35	5	Female	Gold
3	40	6	Female	Gold
4	28	3	Male	Regular
5	50	8	Female	Gold
6	45	7	Male	Silver
7	30	4	Female	Silver
8	55	9	Male	Gold
9	27	3	Male	Regular
10	38	6	Female	Silver

customer_data <- data.frame (

age = c (22, 35, 40, 28, 50, 45, 30, 55, 27, 38),

visits = c (2, 5, 6, 3, 8, 7, 4, 9, 3, 6),

gender = c ("male", "female", "female", "male", "female", "male", "female", "male", "female", "male"),

membership = c ("Regular", "Gold", "Gold", "Regular", "Gold", "Silver", "Silver", "Gold", "Regular", "Silver"),

purchase = c ("Low", "High", "High", "Low", "High", "High", "Medium", "High", "Low", "Medium")

print(customer_data).

Naive Bayes Classifier for Discrete Predictions:

call:

naive Bayes: default ($\mu = \mu$, $\psi = \psi$, Laplace = Laplace)

A-priori probabilities:

Y

	high	low	medium
0.5714286	0.2857143	0.1428571	

conditional probabilities:

	Age		
high	[1, 1]	[1, 2]	
low	0.575	8.539126	
medium	0.450	0.535534	
medium	35.00	NA	
widths			
high	[1, 1]	[1, 2]	
low	6.75	1.7078251	
medium	2.50	0.7071068	
medium	6.00	NA	
Gender			
female			
high	0.5	0.5	
low	0.0	1.0	
medium	1.0	0.0	

- Step 2: Convert Categorical Variables to Factors

```
customer_data$gender <- as.factor(customer_data$gender)
customer_data$membership <- as.factor(customer_data$membership)
customer_data$purchase <- as.factor(customer_data$purchase)
```

- Step 3: Train(70%) - Test Split(30%)

```
index <- sample(1:nrow(customer_data), 0.7 * nrow(customer_data))
train_data <- customer_data[index, ]
test_data <- customer_data[-index, ]
```

- Step 4: Apply Naive Bayes Algorithm

```
#install.packages("e1071") and load
library(e1071)
nb_model <- naiveBayes(purchase ~., data = train_data)
nb_model
```

Membership

	Low	Regular	Silver
High	0.75	0.00	0.25
Low	0.00	1.00	0.00
Medium	0.00	0.00	1.00

→ predicted - purchase

Low High High

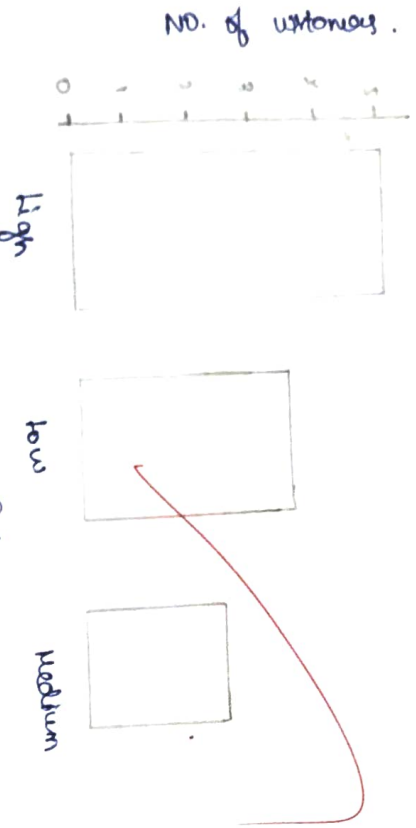
loads: High Low Medium

Confusion - matrix

Predicted

Actual	High	Low	Medium
High	1	0	0
Low	0	1	0
Medium	1	0	0

Purchase Category Distribution



- Step 5: Predict on Test Data

Predicted - purchase & Predict (nb-model, test_data)
Predicted - purchase

- Step 6: Confusion Matrix and Accuracy

confusion-matrix ← table

Actual = test_data \$ purchase

Predicted = predicted - purchase

confusion-matrix

accuracy = sum(diag(confusion-matrix)) / sum(confusion-matrix)

Accuracy

- Step 7: Data Visualization for Naive Bayes Classification

- Class Distribution Plot

- Shows how many customers belong to each purchase category
- Helps identify class imbalance

boxplot (table (customer_data \$ purchase),

col = c("skyblue", "orange", "lightgreen"),

main = "Purchase Category Distribution",

xlab = "Purchase Category",

ylab = "Number of customers")

Customer Distribution by Purchase Category



Age distribution by Purchase Category

Purchase Category



- Scatter Plot (Age vs Visits)
 - Displays customer grouping based on two numerical features - Age and Visits
 - Colors represent purchase categories

plot(customer_data of Age, customer_data of visits,

col = as.numeric(customer_data of purchase),

pch = 19,

main = "Customer Distribution by Purchase Category",

xlab = "Age",

ylab = "Monthly Visits",

legend = "topright",

legend = legend(customer_data of purchase),

col = 1:length(legend(customer_data of purchase)),

pch = 19)

- Boxplot for Numerical Attributes - Age and Purchase Category

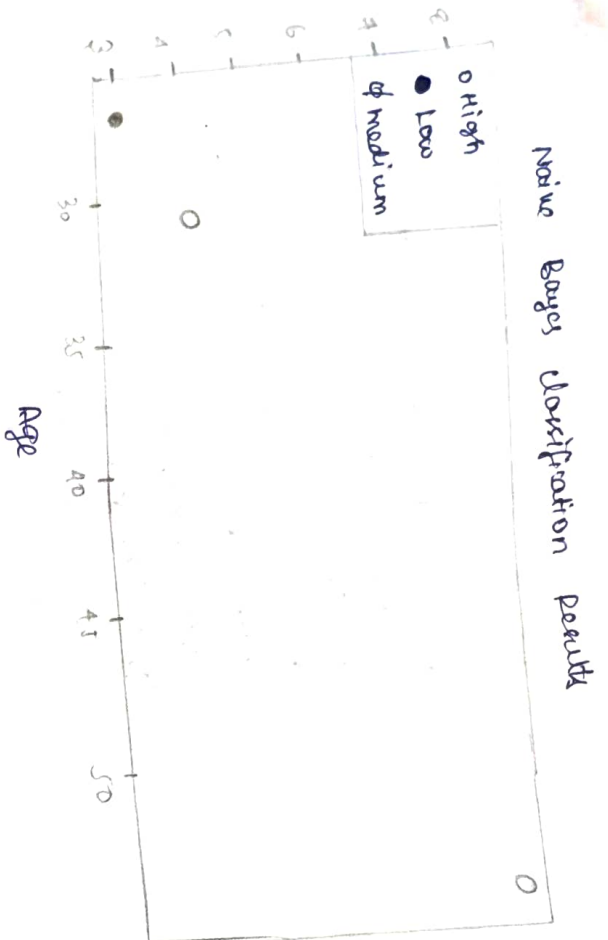
boxplot(Age ~ Purchase, data = customer_data,

col = c("yellow", "blue", "red"),

main = "Age Distribution by Purchase Category",

xlab = "Purchase Category",

ylab = "Age")



Visualization of Prediction Results

```
plot (test_data$Age, test_data$visits,
      col = 0:2, main = "Naive Bayes Classification Results",
      xlab = "Age",
      ylab = "Monthly visits")
legend ("topleft",
        legend = levels (predicted_purchase),
        col = 1:3, length (levels (predicted_purchase)),
        bty = "n")
```

Result:

Thus, the Naive Bayes classifier was successfully implemented in R to predict customer purchase categories using both numerical and categorical attributes. The model effectively applies probabilistic reasoning and produces accurate classification results, making it suitable for customer behavior prediction in e-commerce applications.

> head (iris)

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.8	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.8	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.8	setosa
6	5.4	3.9	1.7	0.4	setosa

Ex: No: 6B
Date: 22/9/20

Iris Flower Species Classification Using Naive Bayes Algorithm

Problem Description:

The Iris dataset contains measurements of iris flowers belonging to three species: **Setosa**, **Versicolor**, and **Virginica**. Each flower is described using four numerical attributes: **Sepal Length**, **Sepal Width**, **Petal Length**, and **Petal Width**. The objective is to classify iris flowers into their respective species using the **Naive Bayes** classifier and visualize the data distribution and classification results.

Objective:

To implement the Naive Bayes classification algorithm in R for predicting iris flower species and to visualize the dataset and classification results.

Procedure:

- Step 1: Load Required Libraries

We load the necessary libraries for implementing the Naive Bayes algorithm.

```
# Install package
install.packages("e1071")
install.packages("caue")
install.packages("ggplot2")

# Load required libraries
library(e1071)
library(caue)
library(ggplot2)
```

- Step 2: Load and Prepare the Dataset

We use the iris dataset for classification and check its structure

```
# Load the iris dataset
data(iris)

# Display first few rows
head(iris)
```

Print (nb-model)

Naive Bayes Classifier for Datasets Prediction

Call:

Naive Bayes, default ($X = X$, $y = y$, Laplace = Laplace)

h-priori probabilities

Setosa Versicolour virginica
0.33333333 0.33333333 0.33333333

conditional probabilities

Setosa

sepal, length
[1, 4]

[1, 8]

Setosa 4.991429 0.3657156
Versicolour 5.942857 0.4553679
virginica 6.631429 0.6846087

sepal, width

[1, 4]

[1, 8]

Setosa 3.365714 0.3455114
Versicolour 4.771429 0.3227508

virginica 4.982857 0.3292035

petal, length

[1, 4]

[1, 8]

Setosa 1.414009 1.008012
Versicolour 4.262857 0.4222389
virginica 5.591409 0.5741344

- Step 3: Split the Dataset into Training and Testing Sets
We split the dataset into 70% training and 30% testing.

Set.seed(123)

train_index <- createDataPartition (iris\$Species, p=0.7,

train_data <- iris[train_index,]

(ytr = FALSE)

test_data <- iris[-train_index,]

- Step 4: Train the Naive Bayes Model

We build the Naive Bayes model using the naiveBayes() function from the e1071 package.

Train Naive Bayes classifier

nb-model <- naiveBayes (Species ~., data = train_data)

Print model details

Print (nb-model)

- Step 5: Make Predictions on Test Data

We use the trained model to make predictions on the test dataset.

Make predictions

predictions <- predict (nb-model, test_data)

Display predictions

Print (predictions)

total width

	L, 0.7	L, 0.7
Setosa	0.2314286	0.09391521
Versicolora	1.3835714	0.20516469
Virginica	2.034285	0.25778371

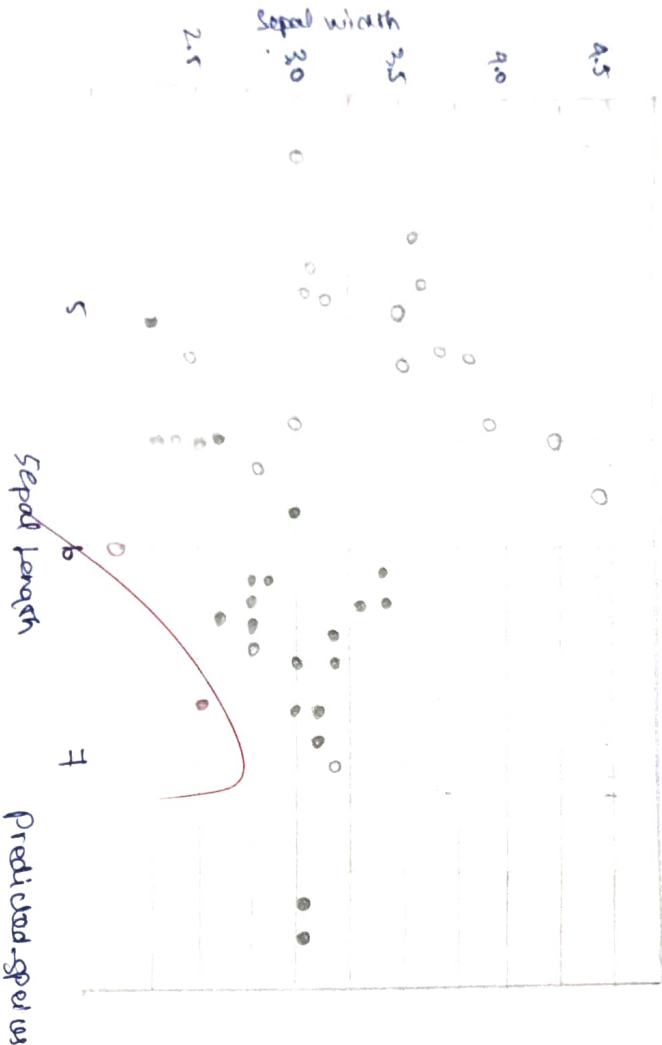
Print (predictions)

[17] setosa	setosa	setosa	setosa	setosa
[18] setosa	setosa	setosa	setosa	setosa
[19] setosa	versicolora	virginica	versicolora	versicolora
[20] virginica	versicolora	versicolora	versicolora	versicolora
[21] versicolora	versicolora	virginica	virginica	virginica
[22] versicolora	versicolora	virginica	virginica	virginica
[23] virginica	virginica	virginica	virginica	virginica
[24] virginica	virginica	virginica	virginica	virginica

Labels: setosa versicolora virginica

Accuracy
0.9111111

Naive Bayes Classification Result



- Step 6: Evaluate Model Performance
We compute accuracy using a confusion matrix.

```
# Confusion matrix
conf_matrix <- confusionMatrix(predictions, test_data$species)
```

```
# Print accuracy
```

```
Print (conf_matrix$overall[1, 'accuracy'])
```

- Step 7: Visualize Classification Results

We plot a scatter plot of Sepal.Length vs Sepal.Width with predicted class labels.

```
# Add predictions to test data
```

```
test_data$Predicted-species <- predictions
```

```
# Scatter plot of Sepal.Length vs Sepal.Width
```

```
ggplot(test_data, aes(x = Sepal.Length, y = Sepal.Width,
```

```
color = Predicted-species))
```

```
geom_point(size = 3))
```

```
ggtitle("Naive Bayes Classification Results")
```

Result:

Thus, the program successfully demonstrates the use of the Naive Bayes algorithm in R to classify iris flower species based on numerical attributes. The visualizations clearly show species separation, and the model achieves high classification accuracy, validating the effectiveness of Naive Bayes for multi-class classification problems.